

Tabular Planning: Policy Evaluation

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CS 1840 Fall 2026

Announcements

HW1 is out today

HW0 is due Sept 17 (**this Thursday**) , 11:59 pm ET

Check the course website for Office Hours:

<https://harvard-cs-1840.github.io/2026/>

Recap: Definitions

$$\mathcal{M} = \{S, A, P, r, \gamma\}$$

$$P : S \times A \mapsto \Delta(S), \quad r : S \times A \rightarrow [0,1], \quad \gamma \in [0,1)$$

Cart pole



$State = \langle$ pole position
 pole angle
 cart angle
 cart velocity $\rangle$

$A = \langle$ left control $\rangle$

Recap: Definitions

$$\mathcal{M} = \{S, A, P, r, \gamma\}$$

$$P : S \times A \mapsto \Delta(S), \quad r : S \times A \rightarrow [0,1], \quad \gamma \in [0,1)$$

Policy $\pi : S \mapsto A$ Deterministic policy

$\Downarrow$

$\pi : S \rightarrow \Delta(A)$ Stochastic policy

Recap: Definitions

$$\mathcal{M} = \{S, A, P, r, \gamma\}$$

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$$\text{Policy } \pi : S \mapsto A$$

Value function $V^\pi(s) = \mathbb{E} \left[\sum_{h=0}^{\infty} \gamma^h r(s_h, a_h) \mid \underline{s_0 = s}, a_h = \pi(s_h), s_{h+1} \sim P(\cdot \mid s_h, a_h) \right]$



Finishing Up From Last Lecture



1. Definitions of Markov Decision Process

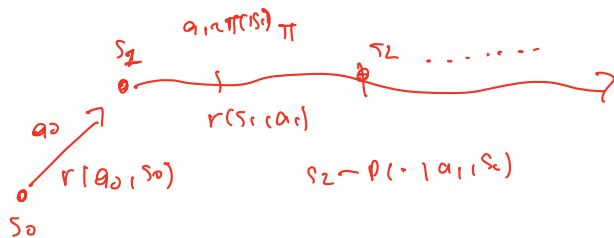
2. Value functions (V and Q functions)

3. Bellman equations

State-action Q function

Δ

$$Q^\pi(s, a) = \mathbb{E} [r_0 + \gamma r_1 + \gamma^2 r_2 + \dots + \gamma^h r_h + \dots \mid \underline{s_0 = s}, \underline{a_0 = a}, \underline{\pi}]$$



Can we express the value function using only one step?

$$V^\pi(s) = \mathbb{E} [r_0 + \gamma r_1 + \gamma^2 r_2 + \dots + \gamma^h r_h + \dots \mid s_0 = s, a \sim \pi]$$

Travel from your Apt to SEC

s_0 - Apt.

s_n - SEC

reward = - travel time

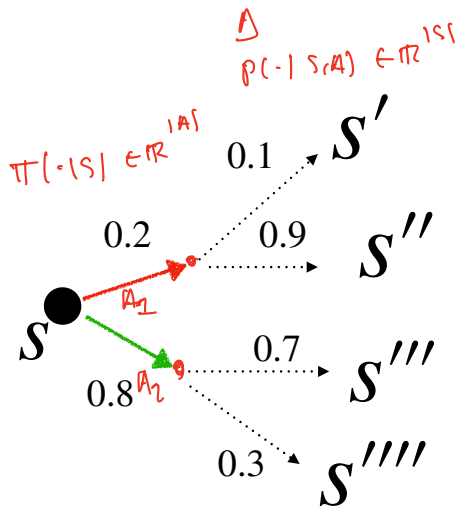


reward = -time (Apt, Bus stop)

$$\begin{aligned} V^\pi(\text{Apt}) &= -\text{time}(\text{Apt}, \text{Bus stop}) \\ &\quad + \gamma V^\pi(\text{Bus stop}) \\ &= -\text{time}(\text{Apt}, \text{Bus stop}) \\ &\quad + \gamma V^\pi(\text{Bus stop}) \end{aligned}$$

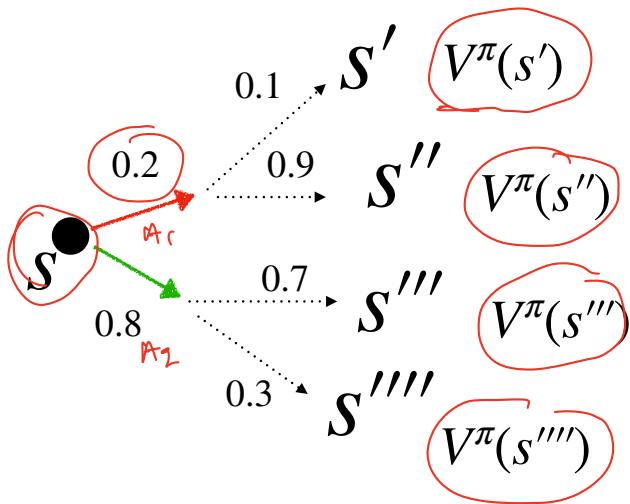
Can we express the value function using only one step?

$$V^\pi(s) = \mathbb{E} [r_0 + \gamma r_1 + \gamma^2 r_2 + \dots + \gamma^h r_h + \dots \mid s_0 = s, a \sim \pi]$$



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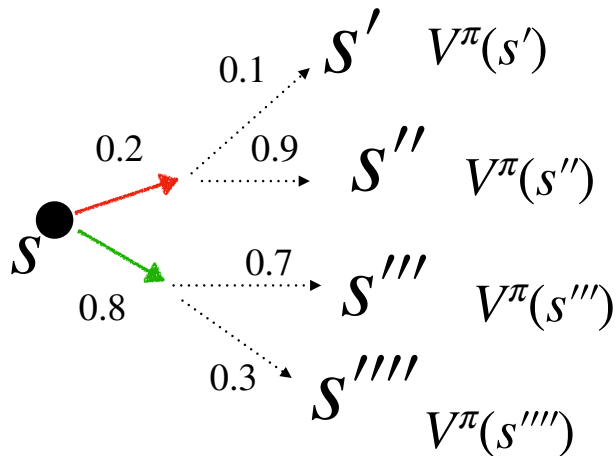
$$V^\pi(s) = \mathbb{E} [r_0 + \gamma r_1 + \gamma^2 r_2 + \dots + \gamma^h r_h + \dots \mid s_0 = s, a \sim \pi]$$



$$\begin{aligned} & \pi(A_1 | s) \times P(s' | s, A_1) \times V^\pi(s') \\ & \quad 0.2 \times 0.1 \times V^\pi(s') \\ & + 0.2 \times 0.9 \times V^\pi(s'') \\ & + 0.8 \times 0.7 \times V^\pi(s''') \\ & + 0.8 \times 0.3 \times V^\pi(s''') \end{aligned}$$

Can we express the value function using only one step?

$$V^\pi(s) = \mathbb{E} [r_0 + \gamma r_1 + \gamma^2 r_2 + \dots + \gamma^h r_h + \dots \mid s_0 = s, a \sim \pi]$$



$$\underline{V^\pi(s)} = \underline{\mathbb{E}_{a \sim \pi(\cdot|s)}} \left[r(s, a) + \gamma \underline{\mathbb{E}_{s' \sim P(\cdot|s, a)}} \left[\underline{V^\pi(s')} \right] \right]$$

Bellman equation for value function

$$\begin{aligned} V^\pi(s) &= \sum_a \pi(a|s) \left[r(s, a) + \gamma \sum_{s'} P(s'|s, a) V^\pi(s') \right] \\ &= \sum_a \pi(a|s) [r(s, a)] + \gamma \sum_a \sum_{s'} \pi(a|s) P(s'|s, a) V^\pi(s') \end{aligned}$$

Bellman equation (consistency)

Bellman Eq

$$V^\pi(s) = \mathbb{E}_{a \sim \pi(\cdot|s)} \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V^\pi(s')] \right], \forall s$$



Relationship between V and Q

Exercise: Can you express V^π in terms of Q^π , and Q^π in terms of V^π ?

$$Q^\pi(s, a) = r(s, a) + \gamma \mathbb{E} [V^\pi(s')]$$

$$V^\pi(s) = \mathbb{E}_{a \sim \pi(a|s)} \left[r(s, a) + \gamma \mathbb{E}_{s' \sim P(s'|s, a)} [V^\pi(s')] \right]$$

$$V^\pi(s) = \mathbb{E}_{a \sim \pi(a|s)} [Q^\pi(s, a)]$$

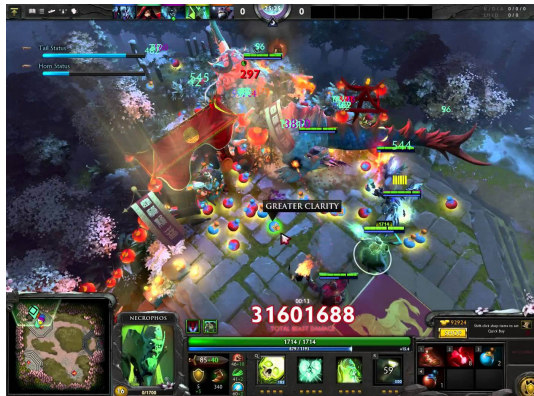
Today: Policy Evaluation

Key Question:

**Given MDP $\mathcal{M} = (S, A, r, P, \gamma)$ and a $\pi : S \mapsto A$,
how good is π ?**

i.e., how to compute $V^\pi(s), \forall s$?

Motivation for Policy Evaluation



We want to **evaluate** our strategy
against some opponent
(we can abstract our strategy as
policy π)

Motivation for Policy Evaluation



We want to **evaluate** our strategy against some opponent (we can abstract our strategy as policy π)



We want to **evaluate** our recommendation strategy before we release it to users

A more fundamental motivation...

Recall that we have A^S many policies.
To select the optimal policy, we need to do evaluation

Outline:

1. **Exact** Policy Evaluation
2. **Approximate** Policy Evaluation via an Iterative Algorithm

Exact Policy Evaluation

Setup: we have MDP $\mathcal{M} = (S, A, P, \gamma, r)$, and deterministic policy π ; we want to compute V^π



Exact Policy Evaluation

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$$= \mathbb{E}_{a \sim \pi} [r(s, a) + \gamma \mathbb{E}_{s' \sim P} [V^\pi(s')]]$$

We know that for V^π , we have **Bellman equation:**

$$a \leftarrow \pi(s)$$

$$\forall s, V^\pi(s) = \boxed{r(s, \pi(s))} + \gamma \mathbb{E}_{s' \sim P(s, \pi(s))} V^\pi(s')$$

Exact Policy Evaluation

Setup: we have MDP $\mathcal{M} = (S, A, P, \gamma, r)$, and deterministic policy π ; we want to compute V^π

We know that for V^π , we have **Bellman equation**:

$$\forall s, V^\pi(s) = r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(s, \pi(s))} V^\pi(s')$$

$V^\pi(s_1) = r(s_1, \pi(s_1)) + \gamma \mathbb{E}_{s' \sim P(s_1, \pi(s_1))} [V^\pi(s')]$

This gives us S many linear constraints

$$V^\pi(s_2) = \dots$$

$$V^\pi(s_3) = \dots$$

Exact Policy Evaluation

Let's form linear constraints. Denote $V(s)$ as our estimator for $s \in S$

state space

$$\forall s, V(s) = r(s, \pi(s)) + \gamma \sum_{s' \in S} P(s' | s, \pi(s)) V(s')$$

Exact Policy Evaluation

Let's form linear constraints. Denote $V(s)$ as our estimator for $s \in S$

$$\forall s, V(s) = r(s, \pi(s)) + \gamma \sum_{s' \in S} P(s' | s, \pi(s)) V(s')$$

Denote $\underline{V} \in \mathbb{R}^{|S|}$, $\underline{R} \in \mathbb{R}^{|S|}$, where $R_s = r(s, \pi(s))$, and
 $P \in \mathbb{R}^{|S| \times |S|}$, where $P_{s,s'} = P(s' | s, \pi(s))$,

$V^\pi = \begin{bmatrix} \bullet \end{bmatrix} \in \mathbb{R}^{|S| \times 1}$ $V(s)$

$R^\pi = \begin{bmatrix} \bullet \end{bmatrix} \in \mathbb{R}^{|S| \times 1}$ $r(s, \pi(s))$

$P^\pi = \begin{bmatrix} \bullet \end{bmatrix} \in \mathbb{R}^{|S| \times |S|}$ $P(s' | s, \pi(s))$

Exact Policy Evaluation

Let's form linear constraints. Denote $V(s)$ as our estimator for $s \in S$

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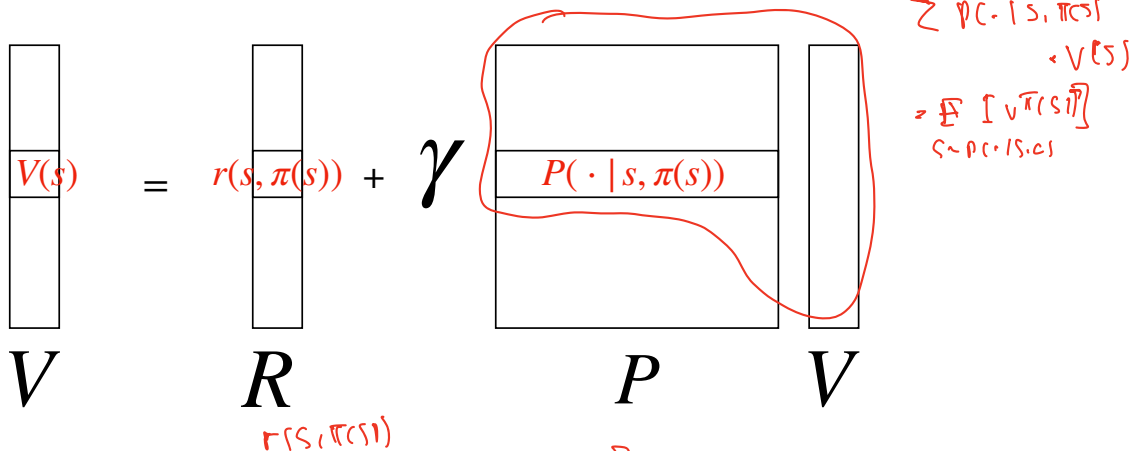
we can **combine all S many constraints together:**

$$\underbrace{V}_{\Delta} = \underbrace{R}_{\Delta} + \gamma \underbrace{P}_{\Delta} \underbrace{V}_{\Delta}$$

Exact Policy Evaluation

$V \in \mathbb{R}^{|S|}$, $R \in \mathbb{R}^{|S|}$, where $R_s = r(s, \pi(s))$, and $P \in \mathbb{R}^{|S| \times |S|}$, where $P_{s',s} = P(s' | s, \pi(s))$,
we can combine all constraints together:

$$V = R + \gamma P V$$



$$V^*(s) = \mathbb{E}_{s \sim \pi} [r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P} [V^*(s')]]$$

Exact Policy Evaluation

Since $V = R + \gamma P V$, we can obtain V as:

$$\Downarrow V = (I - \gamma P)^{-1} R$$

$$V = R + \gamma P V$$

$$V - \gamma P V = R$$

$$V(I - \gamma P) = R$$

$$V = (I - \gamma P)^{-1} R$$

Exact Policy Evaluation

Since $V = R + \gamma PV$, we can obtain V as:

$$V = (I - \gamma P)^{-1}R$$

See the assigned AJKS section for proving $(I - \gamma P)$ is full rank (thus invertible)

Summary so far:

The diagram illustrates the Bellman optimality equation $V = R + \gamma P V$ using rectangular blocks. On the left, a vertical rectangle labeled V has its middle section labeled $V(s)$. This is followed by an equals sign. To the right of the equals sign is a vertical rectangle labeled R with its middle section labeled $r(s, \pi(s))$. This is followed by a plus sign and a large italicized Greek letter γ . To the right of γ is a large rectangle labeled P with its middle section labeled $P(\cdot | s, \pi(s))$. To the right of P is a vertical rectangle labeled V .

$$V = (I - \gamma P)^{-1} R$$

Summary so far:

$$\begin{array}{c} \boxed{} \\ \boxed{V(s)} \\ \boxed{} \end{array} = \begin{array}{c} \boxed{} \\ \boxed{r(s, \pi(s))} \\ \boxed{} \end{array} + \gamma \begin{array}{c} \boxed{} \\ \boxed{P(\cdot | s, \pi(s))} \\ \boxed{} \end{array} \begin{array}{c} \boxed{} \\ \boxed{} \\ \boxed{} \end{array}$$

V
 R
 P
 V

$s \times s$
 $V = (I - \gamma P)^{-1} R$

Downside: computationally expensive; matrix inverse is $O(S^3)$

Outline:



1. Exact Policy Evaluation

2. Approximate Policy Evaluation via an Iterative Algorithm

(An approximation solution could be enough, i.e., trade accuracy for computation)

Detour: fix-point solution

Consider $x^\star = f(\underline{x^\star})$, $f: [a, b] \mapsto [a, b]$

Detour: fix-point solution

Consider $x^\star = f(x^\star)$, $f: [a, b] \mapsto [a, b]$

Common approach to find $x^\star$:

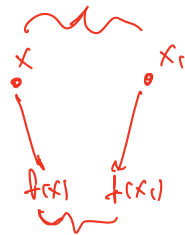
Initialize $x^0 \in [a, b]$, repeat: $x^{t+1} = f(x^t)$ ↗ ~~f(f(f(f(x^0))))~~

Detour: fix-point solution

Consider $x^* = f(x^*)$, $f: [a, b] \mapsto [a, b]$

Common approach to find x^* :

Initialize $x^0 \in [a, b]$, repeat: $x^{t+1} = f(x^t)$



If f is a contraction mapping,

i.e., $\forall x, x', |f(x) - f(x')| \leq \gamma |x - x'|$, for some $\gamma \in [0, 1)$, then:

$x^t \rightarrow x^*$, as $t \rightarrow \infty$

$$\underbrace{|x^{t+1} - x^*|}_{\Delta} = |f(x^t) - f(x^*)| \leq \gamma |x^t - x^*|$$
$$\leq \gamma^2 |x^0 - x^*|$$

V^π is a fix-point solution:

$$\forall s, V^\pi(s) = r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(s, \pi(s))} V^\pi(s')$$



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$$V^\pi = \underbrace{(R + \gamma P V^\pi)}_{f(V^\pi)}$$

Iterative Policy Evaluation:

$$\begin{aligned} & r(s, a) + \gamma V(s) \\ & \gamma \in [0, 1] \\ & 1 + \gamma + \gamma^2 + \gamma^3 + \dots = \frac{1}{1-\gamma} \end{aligned}$$

Algorithm (Iterative PE):

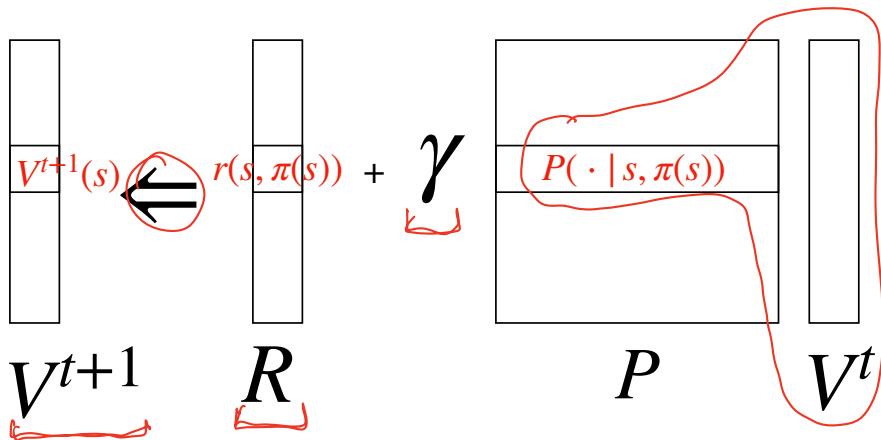
Start with some initialization $\underline{V}^0 \in [0, 1/(1 - \gamma)]^{|S|}$, **repeat for** $t = 0 \dots$:

$$V^{t+1} \leftarrow R + \gamma P V^t$$

Iterative Policy Evaluation:

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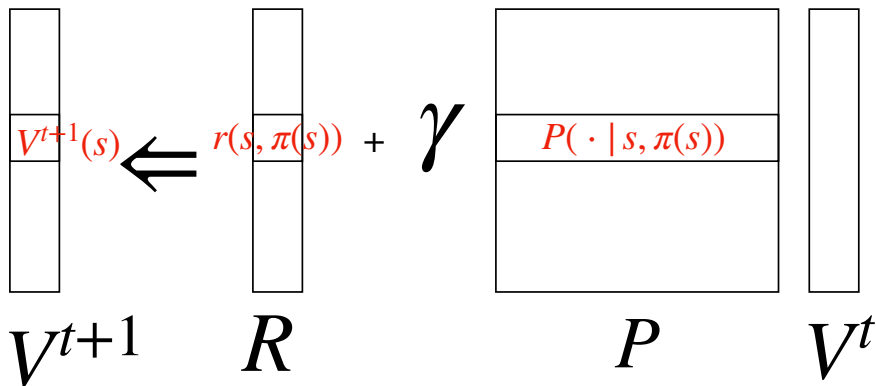
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Iterative Policy Evaluation:

Algorithm (Iterative PE):

Start with some initialization $V^0 \in [0, 1/(1 - \gamma)]^{|S|}$, **repeat for** $t = 0 \dots$:

$$V^{t+1} \leftarrow R + \gamma P V^t$$


Q: What's computation complexity per iteration?

$\mathcal{O}(S^2)$

Iterative Policy Evaluation:

Why does this iterative process makes sense?

$$V^{t+1} \Leftarrow R + \gamma P V^t$$

$$V^{t+1}(s) = r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(s, \pi(s))} V^t(s')$$
$$V^\pi(s) = r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(s, \pi(s))} V^\pi(s')$$

0

$\gamma \cdot \epsilon$

Convergence of Iterative PE

Theorem:

Recall $\gamma \in [0,1)$. After t iterations, we have:

$$\forall s, \left| V^t(s) - V^\pi(s) \right| \leq \gamma^t \left\| V^0 - V^\pi \right\|_\infty$$

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$$= \left| r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, \pi(s))} V^t(s') - \left(r(s, \pi(s)) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, \pi(s))} V^\pi(s') \right) \right|$$

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Convergence of Iterative PE

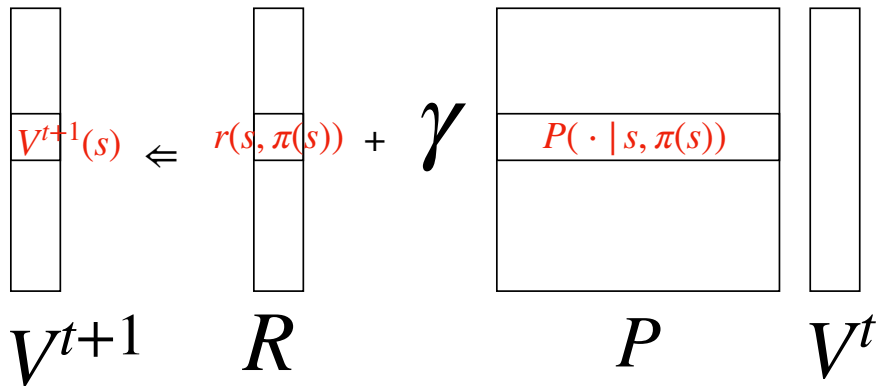
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Summary so far:



Convergence:

$$\| V^{t+1} - V^\pi \|_\infty \leq \gamma \| V^t - V^\pi \|_\infty \leq \gamma^{t+1} \| V^0 - V^\pi \|_\infty$$

Outline:



1. Exact Policy Evaluation



2. Approximate Policy Evaluation via an Iterative Algorithm

Extension to stochastic policy and Q functions

How would we modify exact and iterative policy evaluation to:

1. evaluate a stochastic policy $\pi(a \mid s)$?
2. compute $Q^\pi(s, a)$ instead of $V^\pi(s)$?

Summary

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Key Question today: Given MDP $\mathcal{M}$ and a policy π , How to compute $V(s)$, $\forall s$?

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1. The **exact** algorithm requires the matrix $V = (I - \gamma P)^{-1}R$ inverse (computation complexity at least) $O(S^3)$

Summary

Key Question today: Given MDP $\mathcal{M}$ and a policy π , How to compute $V(s)$, $\forall s$?

1. The **exact** algorithm requires the matrix $V = (I - \gamma P)^{-1}R$ inverse (computation complexity at least) $O(S^3)$
2. An iterative algorithm can quickly find an **approximate** solution (error shrinks at a rate of γ^t)

Summary

Attendance

<https://bit.ly/46pCUdN?r=qr>



Feedback

<https://bit.ly/4gYn2nC>

